

Calibration-aware decision-making with language models		408 works, placed by the stage at which calibration enters
S0 read-off (25 works) Empirical studies of read-off calibration (14) Calibration Across Layers: Understanding... [Joshi et al. 2025] Calibration after alignment (5) Sanz-Guerrero: Large Language... [Sanz-Guerrero et al. 2026] Likelihood and logit read-off (5) My Answer is C: First-Token Probabilities... [Wang et al. 2024d] Self-evaluation and read-off calibration (1) Kadavath: Language Models (Mostly) Know... [Kadavath et al. 2022]	S3 extra inference (121 works) Conformal prediction sets (36) CommnCp: Efficient Multi-Agent Coordination... [Zhang et al. 2026b] Sampling consistency (18) Too Consistent to Detect: A Study of... [Tan et al. 2025] Semantic entropy and meaning clustering (18) Sun: Efficient Hallucination Detection... [Sun et al. 2026a] Risk control and early exit (16) Conformal Thinking: Risk Control for... [Wang et al. 2026b] Conformal abstention, selection and cascades (6) Tayebati: Learning Conformal Abstention... [Tayebati et al. 2025] Conformal factuality and claim filtering (6) Rubin-Toles: Conformal Language... [Rubin-Toles et al. 2025] Benchmarks and empirical studies (5) Vashurin: Benchmarking Uncertainty... [Vashurin et al. 2025] Ensembles and cross-model disagreement (5) Hamidieh: Complementing Self-Consistency... [Hamidieh et al. 2026] Long-form and claim-level (5) IUQ: Interrogative Uncertainty Quantification... [Fan et al. 2026] Internal-state signals (3) Semantic Volume: Quantifying and Detecting... [Li et al. 2026e] Token-level aggregation (3) Word-Sequence Entropy: Towards Uncertainty... [Wang et al. 2025a]	
S1 post-hoc map (52 works) Probes and learned confidence heads (17) Lu: Diagnosing Correctness Probes under... [Lu 2026] Learned calibrators (9) Luo: Unlocking the Pre-Trained Model as a... [Luo et al. 2026] Prompt and context calibration (9) Gundem: Boosting In-Context Learning in... [Gundem et al. 2025] Scaling and binning (7) Luo: Your Pre-trained LLM is Secretly an... [Luo et al. 2025] Fixed maps over the read-off (6) The Silent Vote: Improving Zero-Shot LLM... [Badhe et al. 2026] Label-bias and option-prior debiasing (2) Beyond Performance: Quantifying and... [Reif and Schwartz 2024] Multicalibration and task-specific recalibration (2) Liu: On Calibration of LLM-based Guard Models... [Liu et al. 2024b]	X consumers and contract (71 works) Constrained decoding (19) Reddy: The Hidden Cost of Structured... [Reddy et al. 2026] Routing/cascade (13) Bouchard: Is Escalation Worth It? A... [Bouchard 2026] Structured-output benchmark (10) Singh: The Structured Output Benchmark: A... [Singh et al. 2026a] Human reliance on model output and stated confidence (7) Not All Uncertainty Is Equal: How... [Villavicencio et al. 2026] Adaptive compute (6) Yang: Dynamic Early Exit in Reasoning Models [Yang et al. 2025b] Deferral (4) L2D-Clinical: Learning to Defer for... [Kondadadi and Ortega 2026] Judge calibration (4) JEV-as-a-Judge: Accept When Confident... [Li et al. 2026c] Abstention (3) Srinivasan: Selective "Selective... [Srinivasan et al. 2024] Studies and benchmarks of abstention (3) Wang: Are LLM Decisions Faithful to Verbal... [Wang et al. 2026c] Confidence-weighted selection (1) Taubenfeld: Confidence Improves... [Taubenfeld et al. 2025] Typed classification (1) Look at the Text: Instruction-Tuned... [Wang et al. 2024c]	
S2 prompt elicitation (54 works) Verbalised numeric confidence (30) Yang: The Mirage of Calibrated Confidence... [Yang et al. 2026a] Linguistic hedges and epistemic markers (17) Liu: Can LLMs Use Linguistic Uncertainty... [Liu and Cohan 2026] Introspection, deliberation and task-specific elicitation (3) Reasoning about Uncertainty: Do Reasoning... [Mei et al. 2025] Stated confidence that triggers an action (2) Ferrer: When Can We Trust LLM Graders?... [Ferrer et al. 2026] Abstention prompting (1) I-CALM: Incentivizing Confidence-Aware... [Zong et al. 2026] Self-evaluation prompts (1) Tian: Overconfidence in LLM-as-a-Judge... [Tian et al. 2025]		
Theory and analysis (5 works, not counted in the core) Blasiok: When Does Optimizing a Proper... [Blasiok et al. 2023] Kalai: Calibrated Language Models Must... [Kalai and Vempala 2024] Uncalibrated Reasoning: GRPO Induces... [Bereket and Leskovec 2025] Kalai: Why Language Models Hallucinate [Kalai et al. 2025] Nakkiran: Trained on Tokens, Calibrated... [Nakkiran et al. 2025]		
S4 training: the core (80 works), by what the training signal scores		
T1 supervised calibration objectives (23) Jiang: How Can We Know When Language Models... [Jiang et al. 2021] Chaudhry: Finetuning Language Models to... [Chaudhry et al. 2024] Kapoor: Large Language Models Must Be... [Kapoor et al. 2024] Niu: Functional-level Uncertainty... [Niu et al. 2024] Uncertainty Distillation: Teaching... [Hager et al. 2025] Conf Tuner: Training Large Language Models to... [Li et al. 2025d] Direct Confidence Alignment: Aligning... [Zhang et al. 2025c] Know When You're Wrong: Aligning Confidence... [Xie et al. 2026] T2 reward-model calibration (5) Leng: Taming Overconfidence in LLMs: Reward... [Leng et al. 2025] Lou: Uncertainty-aware Reward Model: Teaching... [Lou et al. 2024] Know What You Don't Know: Uncertainty... [Park et al. 2025a] Ma: Distributional Process Reward Models... [Ma et al. 2026a] Pan: Uncertainty-Aware Reward Modeling for... [Pan et al. 2026a] T5 process- and meaning-level confidence rewards (4) An: Teaching LLMs to Abstain via Fine-Grained... [An and Xu 2025] MMBoundary: Advancing MLLM Knowledge Boundary... [He et al. 2025] Wang: Process Supervision of Confidence... [Wang et al. 2026d] Yu: Calibrating LLMs with Semantic-level Reward [Yu et al. 2026b]	T3 calibration rewards on the stated confidence (31) Band: Linguistic Calibration of Long-Form... [Band et al. 2024] Rewarding Doubt: A Reinforcement... [Bani-Harouni et al. 2026b] Niekerk: Post-Training Large Language... [van Niekerk et al. 2025] Zou: Thinking, Faithful and Stable... [Zou et al. 2025] ORCE: Order-Aware Alignment of Verbalized... [Li et al. 2026a] Singh: Verifiable Rewards for Calibrated... [Singh et al. 2026b] DualStake: Dual-Path Confidence Calibration in... [Xu et al. 2026a] Closing the Reflection Gap: A Free... [Zhu 2026] T4 abstention and refusal training (17) Yang: Alignment for Honesty [Yang et al. 2024b] Chen: Teaching Large Language Models to... [Chen et al. 2025a] Know the Unknown: An Uncertainty-Sensitive... [Li et al. 2025c] Xu: Rejection Improves Reliability: Training... [Xu et al. 2024b] Honesty over Accuracy: Trustworthy... [Mohamadi et al. 2025] TruthRL: Incentivizing Truthful LLMs via... [Wei et al. 2025] TIAR: Trajectory-Informed Advantage... [Pan et al. 2026b] Abstain-RI: Calibrated Abstention and... [Zhao et al. 2026]	

Fig. 2. A taxonomy of calibration-aware decision-making with language models (77 of the 408 works in Tables C1 and F1). Blocks are the stage at which calibration enters (Table A1); X holds the works that consume a confidence or fix the output contract. Every method family at S0 to S3 and X shows its size and its most recent work. The core, S4 (80 works), is grouped by what the training signal scores and lists up to 8 works per family, evenly spaced over the family’s works in year order. The 5 theory and analysis works are listed in full and are not counted in the core. Each entry gives a trimmed form of the paper’s own title; a citation may show a later venue year than the first-version year counted in Figures 6 and 7.